NEPAL COLLEGE OF INFORMATION TECHNOLOGY

SPRING SEMESTR 2025

Level: Bachelor
Program: BEIT
Course: **Data Science and Analytics**

Year: 2025
Full Marks: 100
Pass Marks: 45
Time: 3 hrs.

Candidates are required to answer in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Explain the concept of Knowledge Discovery from Data (KDD) and how it relates to the Data Science Pipeline. 7

   b) Differentiate between structured and unstructured data and Distinguish between numeric and categorical variables. 4+4

2. a) Explain Pearson correlation and how it measures relationship between two numeric variables show it by considering the example. 7

   b) Illustrate the empirical distribution of numeric and categorical data using the following: 8
   1. For numeric: Draw and interpret a histogram. Explain the significance of normal and power-law distributions.
   2. For categorical: Use a bar plot and explain Zipf's law with an example.

3. a) A real estate analyst collected data from 3 houses in a neighborhood to predict house prices based on two features: square footage (in 100s of square feet) and number of bedrooms. 7

Square footage(X1)	Number of bed rooms(X2)	House price (in $1000s) (Y)
15	3	240
18	4	290
20	3	310

Find the regression equation: $Y = \beta 0 + \beta 1X1 + \beta 2X2$

3+5

b) A data analyst builds a multivariate linear regression model to predict house prices based on features such as square footage, number of bedrooms, and distance to the city center.
   1. Explain the significance of R2 and Adjusted R2 in evaluating the performance of the model. Why might Adjusted R2 be a better metric when dealing with multiple predictors?
   2. While analyzing the model, the analyst suspects that some predictors (e.g., square footage and number of rooms) are highly correlated.(Explain in term of multicollinearity and how can it affect the model)
   3. Describe how Variance Inflation Factor (VIF) is used to detect multicollinearity and interpret what high VIF values imply.

4. a) Derive the Nadaraya-Watson kernel regression estimator for the conditional mean E[Y□X=x]. Clearly show how the formula is obtained using kernel weighting .The performance of the Nadaraya-Watson estimator heavily depends on the choice of the bandwidth parameter. Explain the role of the bandwidth in kernel regression and discuss strategies for selecting an optimal bandwidth (e.g., cross-validation, rule-of-thumb methods). 7 8

   b) You are given the following two-dimensional data set consisting of 6 observations:

   (2,1),(3,5),(4,3),(5,6),(6,7),(7,8)
   Using the Principal Component Analysis (PCA) algorithm, compute the first principal component of the dataset

5. a) You are given the following dataset of customers, where the target variable is whether the customer purchased a product (`Yes` or `No`): 7

Age	Income	Purchased
young	high	no
young	medium	yes
middle	high	yes
senior	Low	no
senior	medium	yes
middle	Low	yes

Using the CART (Classification and Regression Tree) algorithm, construct a binary classification tree (up to two levels deep) for predicting whether a customer will purchase the product.

b) Apply the K-Means algorithm with K=2 on the following dataset:(160,55),(175,65),(172,60),(180,70),(178,68),(165,58) Use the first two data points as the initial centroids: and(160,55)and(175,65) Perform the clustering for two iterations, and at each step:

- Assign points to the nearest centroid
- Recompute the centroids
- Show the cluster membership after each iteration

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6. a) What is time series analysis? Explain four components of time series: trend, seasonal variation, cyclical variation and irregular variation. Describe autocorrelation and stationarity with respect to time series analysis. 3+4

b) Explain the differences between the ARMA(Auto-Regressive Moving Average) and ARIMA(Auto-Regressive Integrated Moving Average) models. Define the components of each model:Auto-Regressive (AR) Moving Average(MA) Integration (I). 8

7. Write short notes on: **(Any two)** 2×5
   a) permutation tests, partial and partial dependence plots
   b) Causation vs. correlation
   c) Statistical Significance: p-value
   d) logistic regression